

(a)

Complete second moments

center first: $z \leftarrow z - \langle z \rangle$

ordinary covariance
 $C_{ij} = \langle z_i z_j^* \rangle$

pseudocovariance
 $P_{ik} = \langle z_i z_k \rangle$

proper complex Gaussian: $P = 0$

(b)

three Wick pairings

$$C_{ij}C_{kl}$$

$$C_{il}C_{kj}$$

$$P_{ik}P_{jl}^*$$

omitting P removes one Gaussian contraction

(c)

Connected fourth order

measured fourth moment
 $m_4 = \langle |z|^4 \rangle$



$$R_4^{\text{full}} = \frac{\kappa_4^{\text{conn}}}{v^2} = \frac{m_4}{v^2} - 2 - \frac{|p|^2}{v^2}$$
$$v = \langle |z|^2 \rangle, \quad p = \langle z^2 \rangle$$

independent channels: $\kappa_4^{\text{conn}}(Z) = \sum_{\alpha} w_{\alpha}^4 \kappa_{4,\alpha}^{\text{conn}}$

conditional scaling: $R_4^{\text{full}} \propto N_{\text{eff}}^{-1}$